

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

Chart Construction Test

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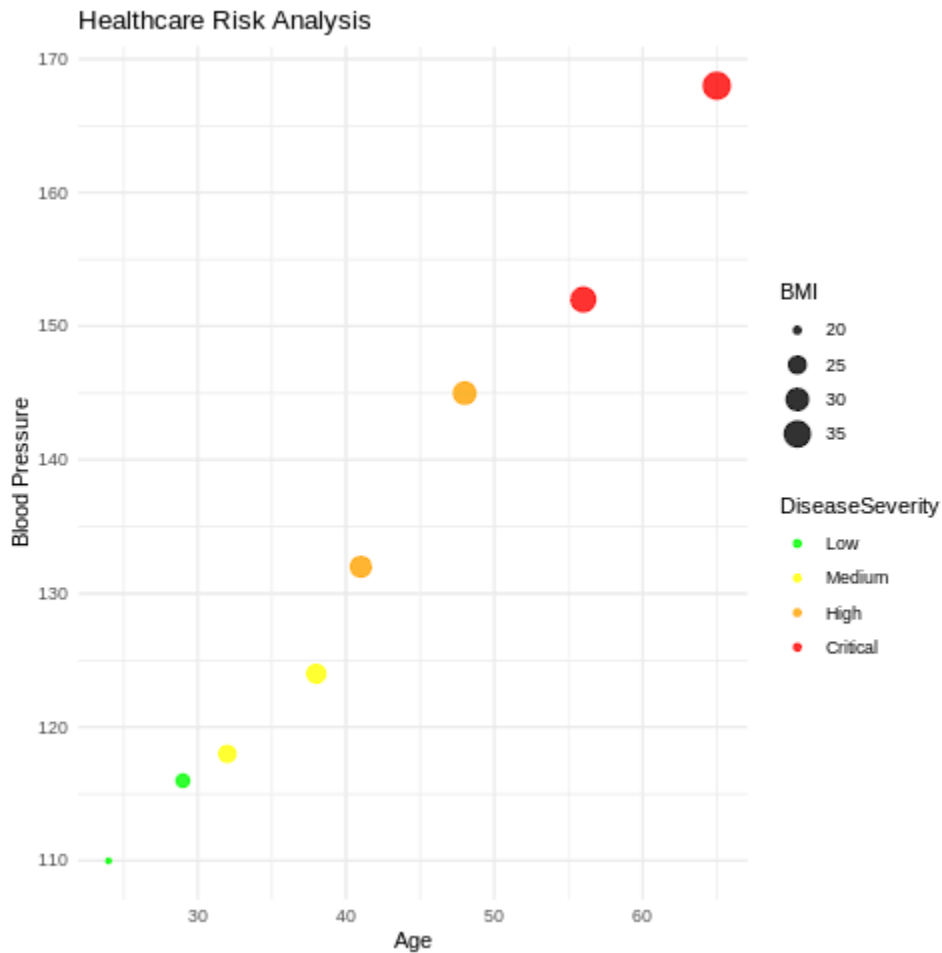
Reg No : 192424098

Question 1 (Healthcare Analytics):

R Code:

```
library(ggplot2)
health <- data.frame(
  Age = c(24,32,41,48,56,65,38,29),
  BMI = c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
  BloodPressure = c(110,118,132,145,152,168,124,116),
  DiseaseSeverity = factor(c("Low","Medium","High","High",
                             "Critical","Critical","Medium","Low"),
                           levels=c("Low","Medium","High","Critical"))
)
ggplot(health,
  aes(x=Age,
      y=BloodPressure,
      color=DiseaseSeverity,
      size=BMI))+
  geom_point(alpha=0.8)+
  scale_color_manual(values=c("green","yellow","orange","red")+
  labs(title="Healthcare Risk Analysis",
      x="Age",
      y="Blood Pressure")+
  theme_minimal()
```

Output:



Justification

- Age → X-axis: Independent health factor.
- Blood Pressure → Y-axis: Indicates cardiovascular risk.
- Disease Severity → Color: Green to Red clearly shows increasing health risk.
- BMI → Point Size: Larger circles represent higher BMI, making obesity easy to identify.
- The visualization helps doctors quickly detect high-risk patients.

Question 2 (Climate Science):

R Code:

Dataset

```
library(ggplot2)
climate <- data.frame(
```

```

Month=factor(month.abb,
levels=month.abb),
SolarRadiation=c(180,200,240,280,320,300,260,250,240,220,190,170),
Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)

```

1. Cartesian Line Chart

```

ggplot(climate,
aes(x=Month,y=Temperature,group=1))+
geom_line(color="blue",size=1.2)+
geom_point(color="red",size=3)+
labs(title="Monthly Temperature Variation",
x="Month",
y="Temperature (°C)")+
theme_minimal()

```

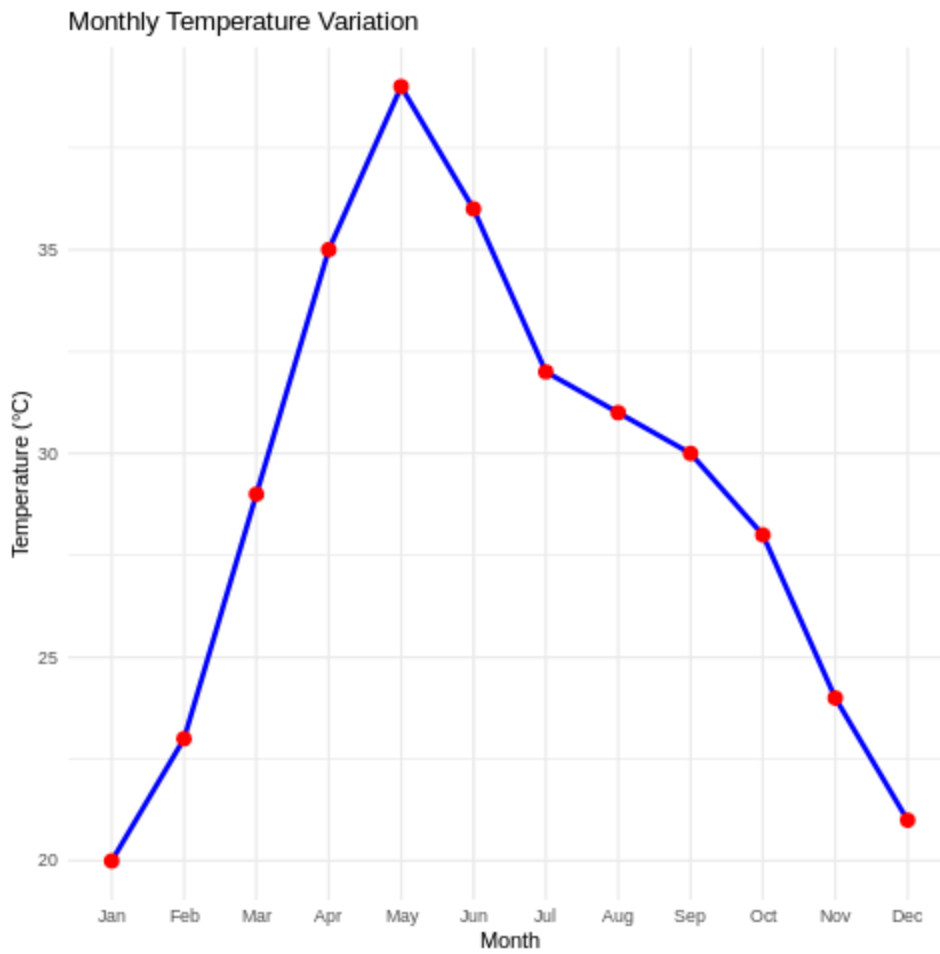
2. Polar Coordinate Chart

```

ggplot(climate,
aes(x=Month,
y=Rainfall,
fill=Rainfall))+
geom_bar(stat="identity")+
coord_polar()+
scale_fill_gradient(low="lightblue",
high="darkblue")+
labs(title="Annual Rainfall Distribution")+
theme_minimal()

```

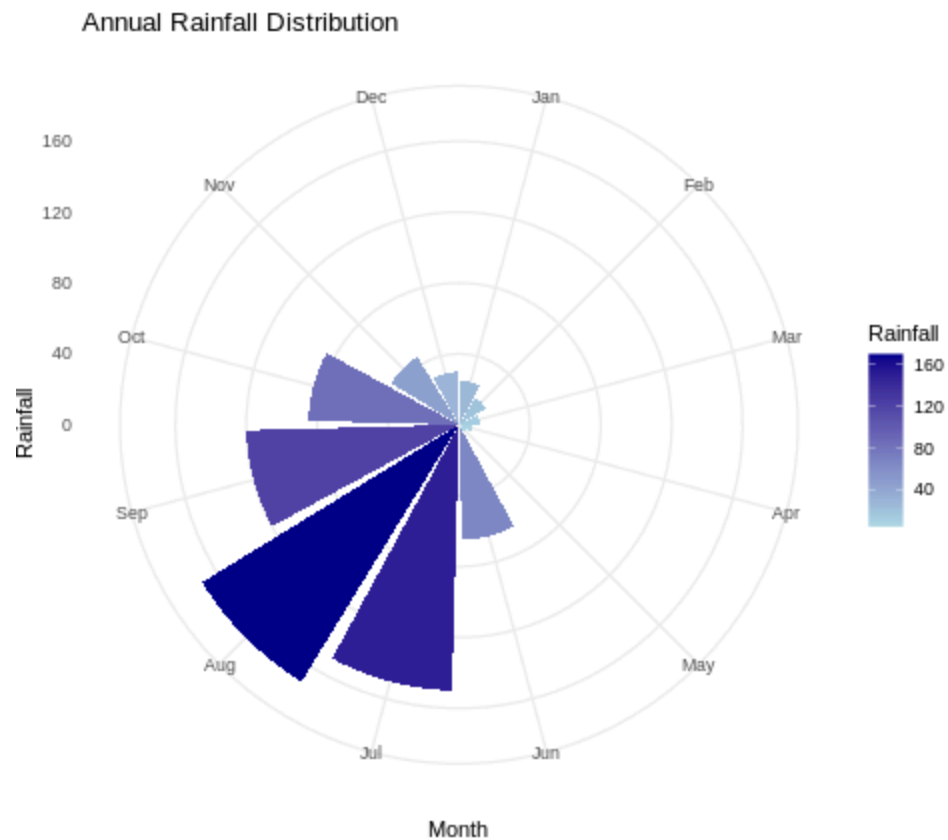
OUTPUT:



2. Polar Coordinate Chart

```
ggplot(climate,  
aes(x=Month,  
y=Rainfall,  
fill=Rainfall))+  
geom_bar(stat="identity")+  
coord_polar()+  
scale_fill_gradient(low="lightblue",  
high="darkblue")+  
labs(title="Annual Rainfall Distribution")+  
theme_minimal()
```

OUTPUT:



Comparison:

Cartesian Chart:

Advantages

- Easy to observe trends.
- Precise value comparison.
- Suitable for continuous variables.

Limitations

- Does not emphasize cyclic nature.

Polar Chart:

Advantages:

- Clearly represents seasonal cycles.

- Attractive visualization for yearly rainfall.

Limitations:

- Harder to compare exact values.
- Small differences may appear distorted.

Question 3 (Business Analytics & Marketing):

R Code :

```
library(ggplot2)
business <- data.frame(
  Product=c("A","B","C","D","E"),
  Sales=c(24,20,28,15,13),
  Profit=c(18,14,25,12,10),
  MarketShare=c(210,185,260,145,120)
)
business$Highlight <- ifelse(
  business$MarketShare==max(business$MarketShare),
  "Highest","Others")
ggplot(business,
  aes(x=Product,
  y=Sales,
  fill=Profit))+
  geom_bar(stat="identity")+
  geom_text(aes(label=Sales),vjust=-0.5)+
  scale_fill_gradient(low="lightgreen",
  high="darkgreen")+
  geom_point(data=subset(business,Highlight=="Highest"),
  aes(x=Product,y=Sales+2),
  color="red",size=5)+
  labs(title="Business Performance Dashboard",
  y="Sales (₹ Lakhs)")+
  theme_minimal()
```

OUTPUT:



Justification

- Bar Height: Represents sales volume clearly.
- Sequential Green Palette: Darker shades indicate higher profit.
- Red Highlight: Draws attention to the highest market share product.
- Helps managers identify both profitability and market leadership quickly.

Question 4 (Transportation Engineering):

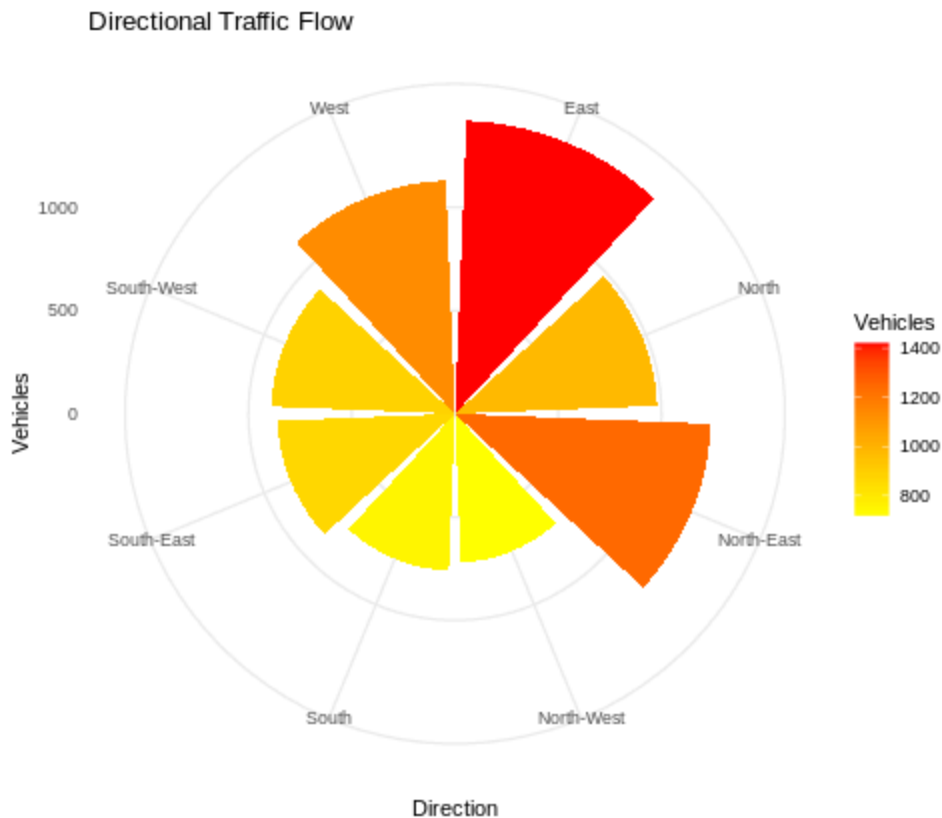
R Code

```
library(ggplot2)

traffic <- data.frame(
  Direction=c("North","North-East","East",
  "South-East","South",
  "South-West","West",
```

```
"North-West"),  
Vehicles=c(980,1240,1420,860,  
760,890,1130,720)  
)  
ggplot(traffic,  
aes(x=Direction,  
y=Vehicles,  
fill=Vehicles))+  
geom_bar(stat="identity")+  
coord_polar()+  
scale_fill_gradient(low="yellow",  
high="red")+  
labs(title="Directional Traffic Flow")+  
theme_minimal()
```

Output:



Justification

- Polar charts naturally represent compass directions.
- Colors distinguish traffic density.
- High-density roads appear in darker colors.
- Better than Cartesian charts because directional data is circular rather than linear.

Question 5(Environmental Sustainability Dashboard):

Dataset

```
library(ggplot2)
```

```
library(gridExtra)
```

```
env <- data.frame(
```

```
City=c("A","B","C","D","E","F"),
```

```
AQI=c(48,82,135,165,72,95),
```

```

GreenCover=c(45,32,21,18,38,30),
CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),
Population=c(9,15,22,28,12,18)
)
env$AQICategory <- cut(
env$AQI,
breaks=c(0,50,100,150,200),
labels=c("Good","Moderate",
"Unhealthy",
"Very Unhealthy")
)

```

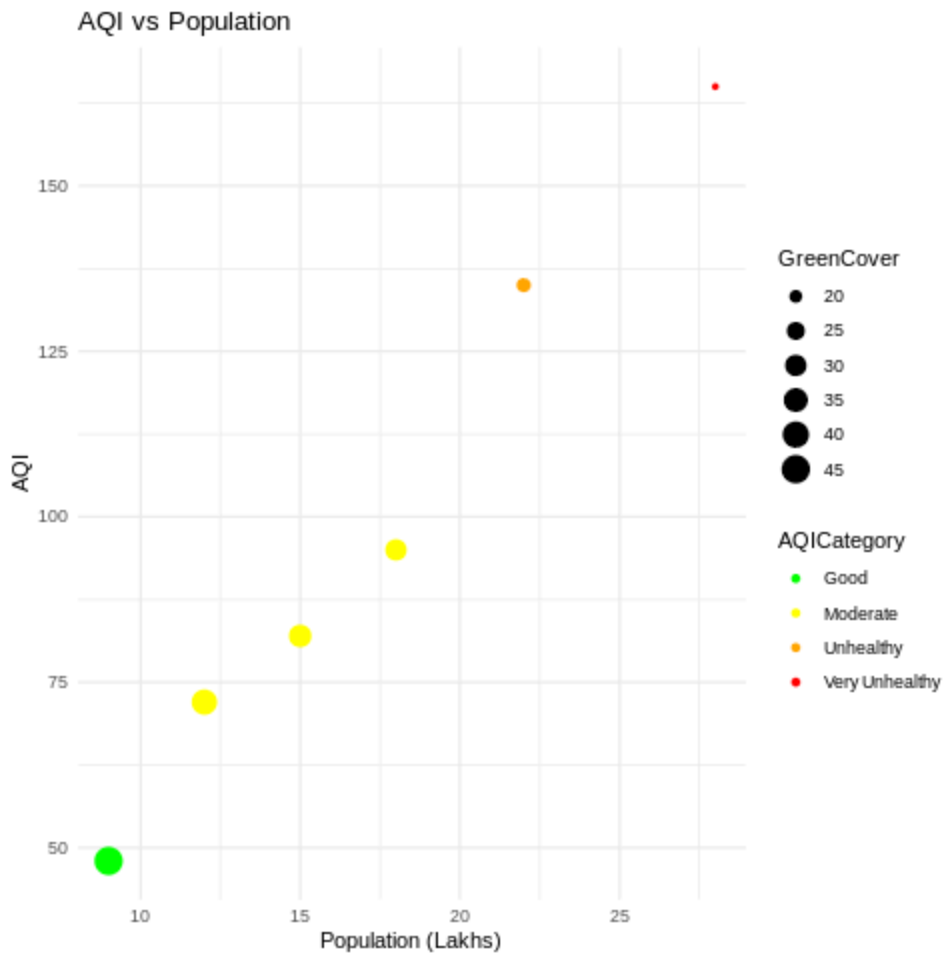
Visualization 1

```

p1 <- ggplot(env,
aes(x=Population,
y=AQI,
color=AQICategory,
size=GreenCover))+
geom_point()+
scale_color_manual(values=c(
"green",
"yellow",
"orange",
"red"))+
labs(title="AQI vs Population")

```

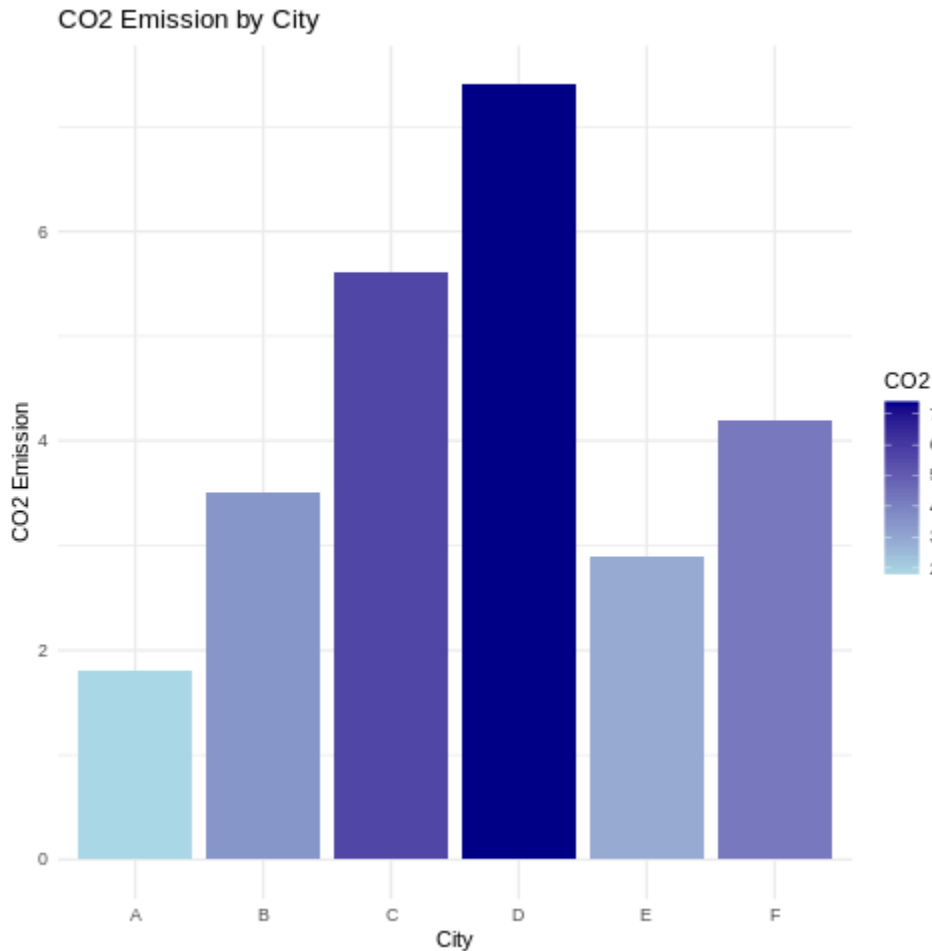
Output:



Visualization 2 :

```
p2 <- ggplot(env,
aes(x=City,
y=CO2,
fill=CO2))+
geom_bar(stat="identity")+
scale_fill_gradient(low="lightblue",
high="darkblue")+
labs(title="CO2 Emission by City")
```

Output:



Design Justification

- Population → X-axis: Represents city size.
- AQI → Y-axis: Measures air quality.
- Green Cover → Point Size: Larger circles indicate more vegetation.
- AQI Category → Color: Green, Yellow, Orange, and Red follow standard air quality categories, making pollution levels easy to interpret.
- CO₂ Bar Chart: Enables quick comparison of emissions across cities.
- Blue Gradient: Darker shades indicate higher CO₂ emissions.
- Dashboard Layout: Displays pollution, population, green cover, and emissions together for a comprehensive sustainability assessment.

